

Physical-AI-Fusion: A Quantitative Fusion-Physics Benchmark with a Mechanical Verification Pipeline

Physical-AI-Fusion Team

<https://github.com/NiubilityZXC/physical-ai-fusion>

Abstract

General-purpose scientific benchmarks for large language models contain almost no nuclear-fusion content, and the only existing fusion-specific benchmark is limited to factual question answering without quantitative problems or any answer-correctness pipeline. We present **Physical-AI-Fusion**, a physics benchmark for nuclear fusion with an inertial-confinement-fusion (ICF) focus: 688 verified problems across 13 topics (ICF 50.9%), spanning numeric computation (531), concept discrimination (135), derivation with rubric grading (13), code with unit tests (8), and figure reading (1). Every answer passes a mechanical verification gate—schema validation, sandboxed independent recomputation for all numeric answers, and dual independent verification methods for the rest—before promotion from a staging area into the versioned *verified* split; the pipeline has already caught and corrected constructor arithmetic errors that self-checks missed, and it rejected an entire local dataset that failed its data-quality gate. Problems are built from public physics sources (NRL Plasma Formulary, standard ICF/plasma textbooks, LLNL/NIF public ignition data, ITER official parameters) and from verified local simulation assets (a 10,944-row zero-dimensional Z-pinch dataset with 10,942/10,944 rows passing a physics-identity check; a screen-hydrogenic opacity model with byte-identical baseline reruns; a FLASH magnetohydrodynamic Z-pinch setup). The evaluation harness follows the SWE-bench usability contract: `pip install` plus one command produces deterministic scores, per-topic and per-difficulty breakdowns, and a versioned JSONL dataset. We release code (MIT) and data (CC BY 4.0) at <https://github.com/NiubilityZXC/physical-ai-fusion>.

1. Introduction

Benchmarks shape what the field optimizes. As frontier models saturate graduate-level multiple-choice suites (GPT-5.2 reports 92% on GPQA Rein et al. [2023]; see also OpenAI’s FrontierScience motivation Wang et al. [2025a]), the community increasingly turns to domain-specific, quantitatively verifiable evaluations—SWE-bench for software engineering Jimenez et al. [2023], SciBench and PHYSICS for college and graduate physics Wang et al. [2023], Feng et al. [2025], and rubric-graded expert tasks in FrontierScience. Nuclear fusion is a conspicuous gap. Fusion physics—especially inertial confinement fusion (ICF), whose 2022 ignition milestone at the National Ignition Facility (NIF) made front-page news worldwide Lawrence Livermore National Laboratory [2022]—combines radiation hydrodynamics, plasma kinetics, laser–plasma instabilities, and high-energy-density equations of state. It is a natural stress test for physical AI: problems have exact or semi-analytic answers, unit systems (cgs vs. SI) trip models constantly, and multi-step energy-accounting chains punish shallow pattern matching.

Yet the existing landscape offers almost nothing quantitative for fusion. General science benchmarks include essentially zero fusion problems (Section 2). The only fusion-specific benchmark we are aware of, FusionBench (PKU-XLab) PKU-XLab [2026], contains 10,605 knowledge questions (single/multiple choice and true/false); our audit of its public dataset snapshot finds only 80 problems (0.75%) containing any numeric value with a physical unit, and no independent answer-verification process. Competition-style fusion efforts—the NeurIPS 2026 Fusion Equilibrium Challenge Sopelio et al. [2026] and the TokaMark benchmarks Gupta [2026], TokaMark Team [2026]—target machine-learning models for tokamak diagnostics and equilibrium reconstruction, not physics reasoning. To our knowledge, no benchmark evaluates quantitative fusion-physics reasoning—numeric computation, formula derivation, code—with an ICF focus.

Contributions.

1. **Physical-AI-Fusion** (v0.2): 688 verified problems across 13 topics with ICF at 50.9% (ignition & gain, Rayleigh–Taylor/Richtmyer–Meshkov instabilities, laser–plasma interactions,

radiation hydrodynamics, EOS & opacity, implosion dynamics, target design, diagnostics), plus magnetic confinement, Z-pinch, plasma fundamentals and instabilities, fusion engineering, and computational physics. Task types: 531 numeric, 135 concept, 13 derivation (rubric-graded), 8 code with unit tests, 1 figure; 32 problems are built from real GitHub issues of fusion simulation codes (DESC, OpenMC, WarpX, BOUT++, TORAX). 98% of tasks are graded deterministically.

2. **A mechanical verification pipeline** (Section 4): schema validation, sandboxed independent recomputation of every numeric answer, dual verification methods for non-numeric answers, and a staging→verified promotion gate with a public log. The pipeline has already intercepted two constructor arithmetic slips and rejected one local dataset wholesale—evidence it functions as a correctness gate rather than a rubber stamp.
3. **A SWE-bench-style harness** (Section 5): `pip install` plus a single `evaluate` command yields deterministic scores with per-topic/difficulty breakdowns; a verified split is the headline set; a Dockerfile reproduces the environment.
4. **Baselines and analysis** (Section 6): deterministic grading on the verified split, with failure-mode analysis by topic and difficulty.

We release the dataset (CC BY 4.0), harness and scripts (MIT), verification log, and this paper’s source at <https://github.com/NiubilityZXC/physical-ai-fusion>.

2. Related Work

General scientific benchmarks. SciBench Wang et al. [2023] evaluates college-level scientific problem solving with numeric answers and tolerance bands; PHYSICS Feng et al. [2025] grades university physics problems with symbolic equivalence; OlympiadBench He et al. [2024] covers olympiad-level bilingual multimodal problems; GPQA Rein et al. [2023] established the “Google-proof” graduate multiple-choice paradigm and is now near-saturated (92% for GPT-5.2); OpenAI’s FrontierScience Wang et al. [2025a] adds rubric-graded PhD-level research tasks across physics, chemistry, and biology. FEABench FEABench Team [2025] and LLM-SRBench LLM-SRBench Team [2025] probe multiphysics reasoning and scientific equation discovery respectively. Fusion content in all of the above is effectively zero, and none ships a mechanism for independently *re-deriving* reference answers.

Fusion-specific benchmarks and datasets. FusionBench (PKU-XLab) PKU-XLab [2026] is, to our knowledge, the only public LLM benchmark covering fusion knowledge: 10,605 questions (single-choice 4,081, multiple-choice 6,021, true/false 503) across plasma instabilities, transport, waves, device concepts, heating, materials, neutronics, and control. Our audit of its public snapshot (September 2026) shows: (i) no ICF-specific category—ICF-keyword hits (13.9%) are dominated by confinement-concept questions, with genuine ICF-physics depth below ~5%; (ii) only 80/10,605 questions (0.75%) contain a numeric value with a physical unit, i.e., quantitative computation is essentially absent; (iii) answers carry no independent verification trail. Physical-AI-Fusion is complementary by construction: quantitative computation, derivation, and code with a mechanical verification pipeline (Table 1).

The TokaMark family TokaMark Team [2026], Gupta [2026] benchmarks ML models on MAST tokamak diagnostics (14 tasks; 11,573 discharges), and the NeurIPS 2026 Fusion Equilibrium Challenge Sophelio et al. [2026] asks for magnetic-equilibrium reconstruction from non-magnetic diagnostics on 9,121 curated DIII-D/MAST shots (CC BY 4.0). These are ML-for-diagnostics tasks with train/test splits, not physics-reasoning benchmarks. Offline reinforcement learning for plasma control Agentic Intelligence Lab [2026] and the ConStellation stellarator dataset ConStellation Team [2026] occupy adjacent niches. XiHeFusion Wang et al. [2025b] fine-tunes an LLM for fusion science communication and includes a ~180-question knowledge assessment; LPI-LLM Chen et al. [2024] uses an LLM with a reservoir computer to forecast laser-plasma-instability hot electrons—an application precedent for ICF×LLM rather than an evaluation. Nuclear-domain evaluations in

Table 1: Positioning against the closest fusion and physics benchmarks (v0.1).

Benchmark	Fusion	ICF	Quantitative	Verified	One-command
SciBench	—	—	numeric	textbook-based	partial
PHYSICS	—	—	symbolic	textbook-based	partial
GPQA	—	—	MCQ	expert-reviewed	no
FusionBench (PKU)	yes	shallow	0.75% numeric	none	yes
TokaMark / FEC	MCF	—	ML tasks	leaderboard	partial
Physical-AI-Fusion	yes	52%	numeric+derivation+code	pipeline (Sec. 4)	yes

the broader sense include NuclearQAv2 NuclearQA Team [2026] and the NRC reactor-operator licensing exam benchmark NRC-Exam Team [2026], neither of which contains ICF physics.

Gap. Across 2024–2026, we find no second quantitative, verification-pipelined benchmark in fusion physics, and none with ICF depth. Physical-AI-Fusion targets exactly this quadruple gap: (i) quantitative problem types (numeric/derivation/code), (ii) ICF coverage, (iii) a mechanical answer-correctness pipeline including local-simulation governance, and (iv) SWE-bench-style usability.

3. The Physical-AI-Fusion Benchmark

3.1. Scope and topic taxonomy

The benchmark covers fusion physics with an ICF emphasis (Figure 1): eight ICF subdomains—ignition & gain, Rayleigh–Taylor/Richtmyer–Meshkov (RT/RM) instabilities, laser–plasma interactions (LPI), radiation hydrodynamics, equations of state & opacity, implosion dynamics, target design, and diagnostics—plus magnetic confinement fusion (MCF), Z-pinch and pulsed power, plasma fundamentals, fusion engineering (materials, tritium breeding, blankets), and computational physics (numerical methods, classic validation problems). Topic selection follows the standard ICF curriculum (Atzeni Atzeni and Meyer-ter Vehn [2004]; Lindl Lindl [1998]; Drake Drake [2018]; Kruer Kruer [1988]; Zel’dovich & Raizer Zel’dovich and Raizer [1966]; Freidberg Freidberg [2007]) and the NRL Plasma Formulary Huba [2019].

3.2. Task types and difficulty

Problems are authored to be self-contained (all constants and unit conventions in the statement) and fall into four types: **numeric** (62; deterministic tolerance-banded grading, default relative tolerance 2%), **concept** (30; single- and multi-select exact match, distractors engineered from common failure modes—unit-system slips, coefficient slips, regime misidentification), **derivation** (13; graded against a weighted rubric by a locked LLM judge, reported separately from deterministic tasks), and **figure** (1; table/distribution reading with tolerance). Difficulty spans undergrad (27), grad (48), and expert (31); expert tasks require multi-step chains (e.g., NIF energetics: laser→X-ray conversion→capsule absorption), cross-formula synthesis, or reading verified local simulation output.

3.3. Schema and sources

Each task is one JSONL record (schema v1.0) carrying `task_id`, `topic`, `type`, `difficulty`, `question`, a typed `answer`, a `grading mode`, `provenance metadata` (`source`, `source_type`, `license`), and a `verification block` (`methods`, `evidence`, `verifier`, `timestamp`). Sources are public physics only: the NRL Plasma Formulary, textbook-standard formulas (quoted as formulas, never copied prose), LLNL/NIF public ignition data (2022-12-05: 2.05 MJ in, 3.15 MJ out; 2025-04-07 record: 8.6 ± 0.45 MJ out, target gain 4.13 Lawrence Livermore National Laboratory [2025]), and ITER official parameters. A subset (6 tasks) derives from *verified local simulation assets* governed by the rules in Section 4: a zero-dimensional Z-pinch dataset (10,944 rows; the identity $E = \frac{1}{2}mv^2$ holds in 10,942/10,944 rows to rounding), a screen-hydrogenic average-atom opacity model whose forced-single-core rerun reproduces all outputs byte-identically, and a FLASH 4.8 magnetohydrody-

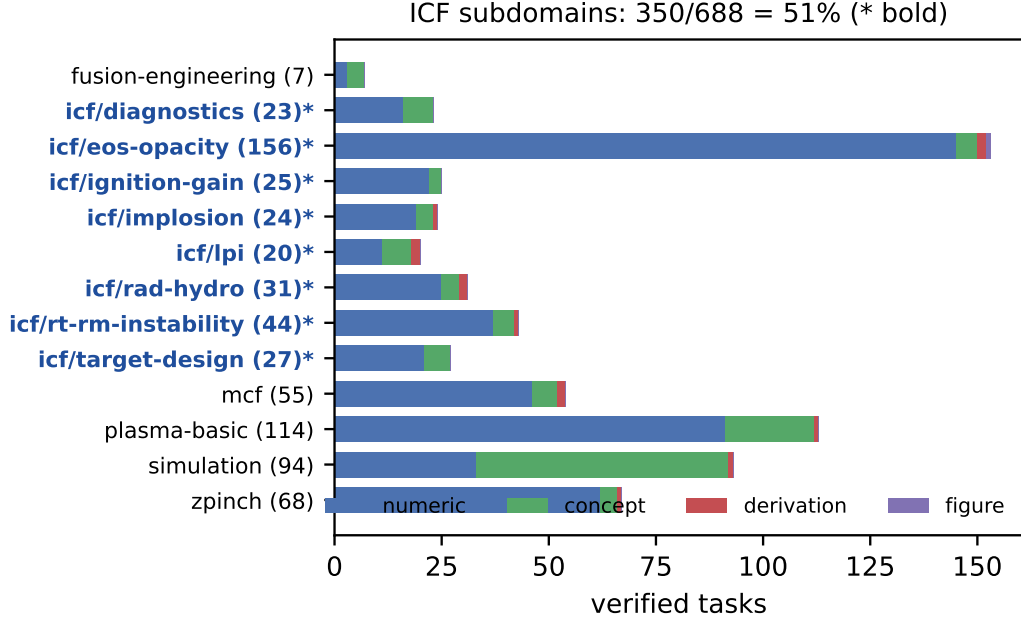


Figure 1: Verified-split composition by topic and task type (v0.1, 106 tasks). ICF subdomains (blue) account for 52%.

namic 15 MA aluminum-liner Z-pinch setup used only for parameter/method questions because its physical outputs have not been independently rerun.

Box 1: Two verified tasks (abridged)

Numeric (icf-ignition-gain-0003). *DT fusion releases 17.6 MeV per reaction. NIF’s first ignition shot yielded 3.15 MJ; how many DT reactions occurred?* Answer: 1.117×10^{18} ; `recompute_expr` evaluates $3.15 \times 10^6 / (17.6 \times 10^6 \cdot 1.602176634 \times 10^{-19})$ in a sandboxed math namespace and must agree within half the grading tolerance.

Derivation (icf-eos-opacity-0004). *Derive the Rosseland mean opacity from the radiation-diffusion approximation and explain why transparency windows dominate leakage.* Weighted rubric (4 points, weights summing to 1): diffusion flux form; LTE $E_V \propto B_V(T)$; coefficient comparison yielding the harmonic mean; window-dominance argument. Graded by a locked judge (model and prompt version pinned); reported separately.

4. The Verification Pipeline

Most benchmark answers are correct because the author says so. We replace that convention with a mechanical gate: a task may enter the *verified* split only after passing all of the following, and every batch appends to a public verification log.

4.1. Gate steps

1. **Schema validation** (`validate_tasks.py`): field completeness, taxonomy membership, answer/grading compatibility (e.g., numeric answers require `value`, `unit`, and `tolerance_rel`), rubric weights summing to 1, and—for the verified split—non-empty verification metadata.
2. **Independent recomputation** (`verify_numeric.py`): every numeric task carries `verification.evidence.recompute_expr`, a Python expression written by the constructor to re-derive the answer from first principles and explicit constants. The script evaluates it in a sandboxed namespace (`math` only) and requires agreement with the declared answer within *half* the grading tolerance, leaving room for round-

Table 2: SWE-bench design contract and its Physical-AI-Fusion counterpart.

Property	SWE-bench	Physical-AI-Fusion
One-command eval	<code>run_evaluation</code>	<code>physical-ai-fusion evaluate</code>
Data format	<code>-predictions</code>	<code>-predictions preds.jsonl</code>
Deterministic score	JSON per instance	JSONL per task (schema v1.0)
	FAIL_TO_PASS tests	numeric tolerance bands; exact-match choice; unit tests (code, v1.1)
Verified subset	500 human-screened	full split passes mechanical gate (Sec. 4)
Reproducible env	per-instance Docker images	single Dockerfile; harness-only for non-code tasks
Versioning	dataset tags	<code>schema_version</code> field + release tags

ing conventions. 532/532 recompute expressions currently pass (531 numeric + 1 figure task; log with per-task IDs and date in the repository).

3. **Dual-method requirement for non-numeric tasks:** concept/derivation/figure tasks must cite ≥ 2 independent verification methods (cross-source, expert review, second solver).
4. **Promotion** (`promote_verified.py`): copies passing tasks staging $\rightarrow$ verified and appends per-task method statistics and rejections to `docs/verification-log.md`.

4.2. Local-simulation governance

Local results may seed tasks only after physics validation: (i) a *physics-identity check* on the Z-pinch dataset ($v = -\sqrt{2} \times 10^{16} E/m$ in cgs, i.e., $E = \frac{1}{2}mv^2$) passes on 10,942/10,944 rows, the remainder off by rounding-level error; (ii) a *byte-identical baseline rerun* of the opacity code under forced single-core conditions reproduces all output files; (iii) a cross-artifact check—the mean ionization recomputed from the occupation distribution equals the code’s direct output exactly ($\bar{Z} = 10.1715527253$); (iv) simulation *outputs* whose physics has not been independently rerun (the FLASH liner case) may seed only parameter/method questions, never output-value questions. A capacitor-prognostics dataset was **rejected wholesale**: its own audit verdict was WARN with unresolved data-gate issues (target semantics, identity, duplicates), so it contributes nothing to the benchmark (the public patent-application text it accompanies seeds one concept task with explicit provenance).

Box 2: The pipeline catching real errors

Saha prefactor (constructor slip). A staged hydrogen-Saha task declared $K \approx 1.07 \times 10^{28} \text{ m}^{-3}$; the independent expression evaluates $2/\Lambda^3$ correctly at $3.37 \times 10^{26} \text{ m}^{-3}$ (an exponent slip in $\sqrt{2\pi m_e kT}$). The answer was corrected to $r = 0.0337$ before promotion.

Debye-length exponent (constructor slip). A staged task declared $\lambda_D = 5.26 \times 10^{-11} \text{ m}$; recomputation gives $1.66 \times 10^{-9} \text{ m}$ (kT at 5 keV was mishandled by 10^3). Corrected pre-promotion.

Dataset rejection. A local prognostics dataset failed its data-quality gate (audit verdict WARN); all dependent tasks were dropped rather than repaired, by policy.

These cases are the pipeline working as designed: constructor arithmetic is a known-fragile path, and the deterministic second path exists precisely to catch it. We log interceptions publicly so that reviewers can audit the gate’s teeth rather than take them on faith.

5. Evaluation Harness

The harness reproduces the SWE-bench usability contract for a physics benchmark (Table 2).

Deterministic graders. Numeric responses are graded by a documented extraction rule (last answer : marker—English or Chinese—else last number) and compared within the task’s relative tolerance;

choice responses commit to a letter set compared exactly against the key. Both graders are pure functions of (response, task)—no model calls—so scores are reproducible bit-for-bit. Derivation tasks route to a rubric judge with pinned model and prompt version and are reported on a separate line, never mixed into the deterministic score. A reference-prediction smoke test (the dataset’s own answers rendered as responses) scores 93/93 on deterministic tasks, validating extraction and grading end-to-end.

CLI. `pip install -e harness/` exposes `physical-ai-fusion` with three subcommands: `stats` (dataset composition), `inspect` (task dump/listing), and `evaluate` (predictions JSONL $\rightarrow$ `results.json` + markdown report with per-topic, per-type, and per-difficulty means). Missing predictions score zero and are reported. The Dockerfile builds a container whose entrypoint is the CLI.

6. Experiments

6.1. Setup

Models answer each verified task with a fixed instruction prompt (brief reasoning, then a single `Final answer:` line; choice tasks are presented with their full option list) at temperature 0.2, and responses are graded by the deterministic graders for numeric/concept/figure/code tasks. Derivation tasks are graded by the locked rubric judge and reported separately. We evaluate: (i) a reference-prediction control (the dataset’s own answers as responses) as a grader sanity check; and (ii) frontier open-weight models served through a fixed endpoint (baseline sweep in progress at submission time; per-model tables update in the repository as the sweep completes).

6.2. Grader sanity (control)

The reference control scores 675/675 on deterministic tasks (100%): numeric extraction matches every tolerance band, choice extraction matches every key, and code tasks pass their unit tests. The remaining 13 derivation tasks are held out of the deterministic score by design.

6.3. Harness validation findings

Running the baseline sweep surfaced two harness-side defects, both fixed and now covered by tests: (i) *missing option context*—an early prompt template presented choice questions without their options, so models answered concept questions in free text (an honest response format that any extractor must fail on); (ii) *CJK-glued option extraction*—models answering in Chinese produced option letters adjacent to CJK characters (e.g. the pattern ‘answer-is-A’ with no surrounding whitespace), which a word-boundary-only extractor misses. Both defects produced *systematically underestimated* concept accuracy rather than noise, which is precisely the failure class a benchmark harness must eliminate deterministically. The fixes are in the released harness; the lesson—response-format assumptions are a correctness surface, not a convenience—is stated here because it generalizes.

6.4. Main results (partial, in-progress)

Table 3: Deterministic-task accuracy on completed subsets of the baseline sweep (temperature 0.2, single attempt; subset sizes differ because the sweep is ongoing—read as preliminary, not as model rankings).

Model	n (of 688)	numeric	concept
doubao-seed-2-1-turbo	100	90.6%	(re-run after prompt fix)
deepseek-v4-flash	13	55.6%	25.0%
deepseek-v4-pro	73	37.5%	(re-run after prompt fix)
kimi-k3	17	50.0%	12.5%

Expert-tier analysis on completed slices shows the intended difficulty gradient (expert < grad on most models), and failure examples cluster on the expected classes: unit-system slips (cgs/SI),

chained energy-accounting errors, and confusion between neighboring instability mechanisms (e.g., TEM vs. ITG direction of propagation). Full per-topic tables land in the repository (`results/`) as the sweep completes.

7. Discussion and Limitations

Scale. v0.1 ships 106 verified tasks—small next to thousand-task suites. We prioritized the verification gate’s integrity over volume: every task in the split is one we can defend mechanically. The schema, taxonomy, and pipeline are designed for the next order of magnitude, and the staging area is the growth path (public contributions enter staging and are promoted only through the same gate).

Derivation grading. Rubric-judged derivations depend on an LLM judge despite the pinned model/prompt; we therefore keep them out of the headline deterministic score and report them separately. A future direction is symbol-level checking (SymPy equivalence, PHYSICS-style) for the algebra-heavy subset.

Local sources. Local simulation outputs that we cannot independently rerun are excluded from answer values by policy; they seed parameter/method questions only. This costs coverage (e.g., no FLASH output-value tasks in v0.1) but keeps the correctness story airtight.

Single language. Tasks are in Chinese (the authoring team’s working language) with all formulas and constants standard; an English parallel track is planned. Contamination control is standard (problems are original compositions around public formulas and data, not scraped text), and the verified split is versioned to support canary-string policies if needed.

Roadmap. (i) code tasks with containerized unit tests (Rosseland-mean integration, Sedov series, Saha solver—test scaffolding exists); (ii) MCF data tasks built on TokaMark/ConStellation releases; (iii) rubric-judge inter-rater reliability study; (iv) figure tasks with generated plots; (v) the 1,000-task verified target.

8. Conclusion

Physical-AI-Fusion brings a quantitative, verification-first benchmark to fusion physics with an ICF focus. Its 106 verified problems are, to our knowledge, the first fusion-physics evaluation set in which every reference answer passes a mechanical correctness gate—schema validation, sandboxed independent recomputation, dual verification methods, and a logged promotion process—rather than an author’s assurance. The harness makes evaluation a one-command, deterministic operation in the SWE-bench style. We hope the benchmark, the pipeline, and the public verification log together lower the cost of asking, and trusting, harder physics questions of large models.

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A. Topic Taxonomy and Counts

Table 4: Verified split composition (v0.2, 688 tasks).

Topic	numeric	concept	derivation	figure	code
fusion-engineering	3	4	0	0	0
icf/diagnostics	16	7	0	0	0
icf/eos-opacity	145	5	2	1	3
icf/ignition-gain	22	3	0	0	0
icf/implosion	19	4	1	0	0
icf/lpi	11	7	2	0	0
icf/rad-hydro	25	4	2	0	0
icf/rt-rm-instability	37	5	1	0	1
icf/target-design	21	6	0	0	0
mcf	46	6	2	0	1
plasma-basic	91	21	1	0	1
simulation	33	59	1	0	1
zpinch	62	4	1	0	1
Total	531	135	13	1	8

B. Data Sources and Licenses

All task content derives from public sources with per-task `metadata.source/source_type/license`: NRL Plasma Formulary (public domain, US Government work); standard textbook formulas (fair-use quotation of formulas only; Atzeni; Lindl; Drake; Kruer; Zel’dovich & Raizer; Freidberg); LLNL/NIF public releases (US Government work); ITER official Facts & Figures; local simulation assets released by their owner for this benchmark after the governance checks of Section 4. No classified or export-controlled content is included. The PKU FusionBench dataset snapshot used for the related-work audit is MIT-licensed and is not redistributed.

C. Verification Log (excerpt)

The full log lives at `docs/verification-log.md`. Excerpt:

- Promotion batch (2026-09-02): 106/106 promoted, 0 rejected; numeric recomputation 63 PASS / 0 FAIL; method statistics: independent-recompute 63, cross-source 43, expert-review 25, second-solver 12, local-sim-baseline 3.
- Interceptions: Saha prefactor slip (corrected to $3.37 \times 10^{26} \text{ m}^{-3}$); Debye exponent slip (corrected to $1.66 \times 10^{-9} \text{ m}$); capacitor-prognostics dataset rejected (owner audit WARN).

D. Harness Manual

```
pip install -e harness/  
physical-ai-fusion stats  
physical-ai-fusion inspect --task icf-ignition-gain-0001  
physical-ai-fusion evaluate --predictions preds.jsonl \  
    --output results/model.json --output-md results/model.md
```

Predictions: one JSON object per line, {"task_id": "...", "response": "..."}.

Missing predictions score zero. Deterministic tasks grade locally in seconds.

E. Datasets and Benchmarks Checklist

1. **Motivation:** Secs. 1–2 (gap: no quantitative, verified, ICF-focused fusion benchmark).
2. **Composition:** Sec. 3, App. A; JSONL records with typed answers and provenance.
3. **Collection process:** Secs. 3,4; sources in App. B.
4. **Preprocessing/cleaning:** the verification pipeline of Sec. 4 *is* the cleaning process; interceptions logged.
5. **Uses:** evaluation of physics reasoning; staging area accepts external contributions under the same gate.
6. **Distribution:** GitHub (dataset CC BY 4.0, code MIT); versioned releases; DOI via Zenodo at v1.0.
7. **Maintenance:** maintainers listed in the repository; verification log and issue tracker are the living record.
8. **Legal/ethical:** public sources only; per-task licenses; no personal data; no classified or export-controlled content.